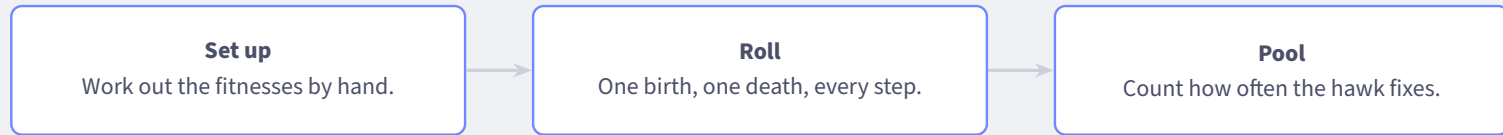


Moran process

Three individuals, two dice, one survivor



Pairs playing:

Starting state: one hawk, two doves

	Hawk	Dove
Hawk	0	3
Dove	1	2

Four resources to share. Two hawks get each; a hawk meeting a dove takes to the dove's ; two doves split them each.



The number of hawks. Along the top a hawk is born; along the bottom one dies. The two ends are absorbing, so the process stops only once one type is gone.

A type's fitness counts what it scores against everyone *else* in the population, so it never plays itself.

State	$f(\text{hawk})$	$f(\text{dove})$
1 hawk, 2 doves		
2 hawks, 1 dove		

Which type is fitter when hawks are rare?

And when they are common?

Birth is proportional to fitness. Death is uniform, so it is proportional to _____.

State	Type	Chosen for birth	Chosen for death
1 hawk, 2 doves	hawk		
1 hawk, 2 doves	dove		
2 hawks, 1 dove	hawk		
2 hawks, 1 dove	dove		

These are the numbers the dice have to reproduce, which is why we need a four and a six.

The number of hawks changes only when the type born differs from the type that dies.

$$p_{10} =$$
$$p_{12} =$$
 $p_{21} =$ $p_{23} =$

Go back and write these on the chain. What happens with the remaining probability?

Runs that ended with all hawks: out of , so a fixation probability of about .

The theory says $x_1 = \frac{1}{1 + \gamma_1 + \gamma_1 \gamma_2}$ with $\gamma_i =$, giving .

Close enough? How many runs would we need to be sure?

What we just did in the room is **Question 1** on the Moran Process page, with a population of four instead of three.

- (a) The two random choices made at each step. [3]
- (b) Fitnesses with two hawks and two doves. [6]
- (c) The probability the number of hawks increases by one. [7]
- (d) Define fixation, and say why there are exactly two absorbing states. [5]
- (e) The probability it decreases, the ratio γ , and which way it drifts. [4]

vknights.org/gt/topics/moran-process.html